Land 15: Genesis Gardens

Reproduction and development · OpenStax Ch. 27 and 28

Key ideas

Male reproductive anatomy

- Testes sit in the scrotum, which keeps them a few degrees cooler than core body temperature, as spermatogenesis requires.
- Seminiferous tubules produce sperm; Sertoli cells nourish germ cells and form the blood testis barrier; Leydig (interstitial) cells between tubules secrete testosterone.
- Sperm path: seminiferous tubules, rete testis, efferent ductules, epididymis (maturation and motility), ductus deferens, ejaculatory duct, urethra.
- Seminal vesicles supply about 60 percent of semen volume with fructose; the prostate adds a milky fluid with PSA and clotting enzymes; bulbourethral glands add a thick lubricating fluid.
- Clinical: cryptorchidism impairs spermatogenesis; benign prostatic hyperplasia compresses the urethra; vasectomy cuts the ductus deferens.

Spermatogenesis and the male HPG axis

- Sequence: spermatogonium (2n, mitosis), primary spermatocyte (meiosis I), secondary spermatocyte (meiosis II), spermatid, sperm.
- One primary spermatocyte yields four spermatids. Spermiogenesis remodels spermatids into sperm with head (acrosome), midpiece (mitochondria) and flagellum.
- Hypothalamus releases GnRH; the anterior pituitary releases FSH (to Sertoli cells) and LH (to Leydig cells).
- Testosterone feeds back to lower GnRH and LH; inhibin from Sertoli cells lowers GnRH and FSH when sperm count is high.
- Exogenous anabolic steroids suppress GnRH, LH and FSH, causing testicular shrinkage and low sperm count.

Female reproductive anatomy

- Ovaries hold follicles and secrete estrogen and progesterone.
- Uterine tubes have fimbriae that sweep in the oocyte, a wide ampulla (usual site of fertilization) and a narrow isthmus; cilia and smooth muscle move the oocyte.
- Uterine wall layers: perimetrium (outer serosa), myometrium (smooth muscle), endometrium (stratum functionalis shed in menses, stratum basalis regenerates it).
- The vagina is kept acidic by Lactobacillus producing lactic acid, which discourages pathogens.
- Mammary glands: milk is made in alveoli, travels through lactiferous ducts and collects in lactiferous sinuses near the nipple.

Meiosis, oogenesis and follicles

- Mitosis: one division, two identical diploid cells. Meiosis: two divisions, four genetically distinct haploid cells; crossing over and independent assortment add variety.
- Primary oocytes form before birth and stay arrested in prophase I until puberty.
- Each cycle a primary oocyte finishes meiosis I, forming a secondary oocyte and a first polar body; the secondary oocyte arrests in metaphase II.
- Meiosis II finishes only if a sperm penetrates the oocyte, so one primary oocyte yields one ovum plus polar bodies.
- Follicle sequence: primordial, primary, secondary, tertiary (antral); the dominant tertiary follicle ovulates.

Ovarian and uterine cycles

- Ovarian cycle: follicular phase (FSH drives follicle growth, estrogen rises), ovulation, luteal phase (corpus luteum secretes progesterone and estrogen).
- Sustained high estrogen switches feedback from negative to positive, causing the LH surge that triggers ovulation.
- Uterine cycle: menses (functionalis shed), proliferative phase (estrogen rebuilds the lining), secretory phase (progesterone readies glands and spiral arteries).
- Without pregnancy the corpus luteum becomes a corpus albicans, progesterone falls, and menses begins.
- Menopause: follicles run out, estrogen and progesterone fall, and FSH and LH rise from loss of negative feedback.
- Endometriosis: endometrial tissue outside the uterus responds to cycle hormones and causes cyclic pain.

Fertilization to implantation

- Sperm push through the corona radiata, bind the zona pellucida, and the acrosomal reaction releases enzymes that digest a path.
- Fusion of one sperm triggers the cortical reaction, which alters the zona pellucida and blocks polyspermy.
- Cleavage: rapid mitotic divisions without growth form blastomeres, then a morula, then a blastocyst.
- Blastocyst: inner cell mass becomes the embryo; trophoblast forms the chorion and fetal placenta.
- The blastocyst implants in the endometrium about a week after fertilization; the trophoblast secretes hCG, which keeps the corpus luteum alive.
- Pregnancy tests detect hCG; an ectopic pregnancy, usually in the uterine tube, still makes hCG and risks tubal rupture.

Embryonic and fetal development

- Gastrulation forms three germ layers: ectoderm (epidermis, nervous system), mesoderm (muscle, bone, blood, heart, kidneys), endoderm (gut and respiratory linings, liver, pancreas).
- The notochord (mesoderm) induces the neural tube and later remains as the nucleus pulposus.
- Embryonic period: fertilization through week 8, when organogenesis lays down all organ systems; fetal period: week 9 to birth, mainly growth and maturation.
- The placenta forms from the chorion plus maternal endometrium and takes over progesterone and estrogen production.
- Umbilical cord: two umbilical arteries carry oxygen poor blood to the placenta; one umbilical vein carries oxygen rich blood to the fetus.

Pregnancy, labor, lactation and birth

- Maternal changes include increased blood volume, cardiac output and respiratory demand as the uterus and placenta grow.
- Labor stages: dilation of the cervix, expulsion of the newborn, afterbirth (delivery of the placenta).
- Oxytocin and cervical stretch form a positive feedback loop that intensifies contractions until birth.
- Prolactin (anterior pituitary) drives milk synthesis; oxytocin (posterior pituitary) drives milk ejection; suckling stimulates both.
- At birth: foramen ovale closes (fossa ovalis) as left atrial pressure rises; ductus arteriosus becomes the ligamentum arteriosum; ductus venosus becomes the ligamentum venosum.

Inheritance

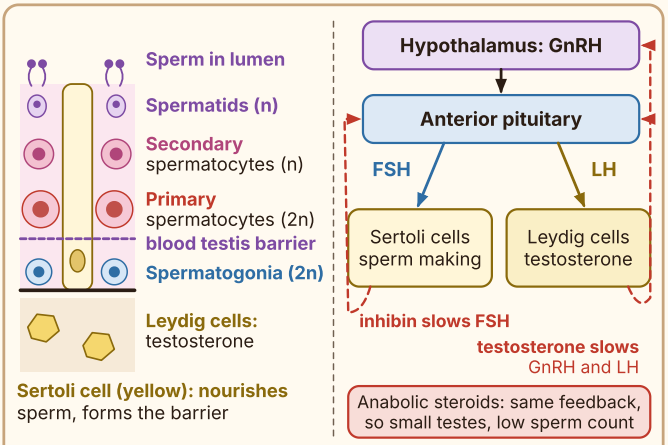
- Humans have 22 pairs of autosomes and one pair of sex chromosomes (XX or XY).
- Genotype is the allele combination; phenotype is the observable trait. Homozygous means two identical alleles; heterozygous means two different alleles.
- A dominant allele is expressed in heterozygotes; a recessive trait shows only when both alleles are recessive.
- Punnett squares: Aa x Aa gives 1 AA, 2 Aa, 1 aa (25 percent recessive phenotype); Aa x aa gives 50 percent recessive phenotype.
- X linked recessive traits (color blindness, hemophilia) are more common in males, who have one X; a carrier mother passes the allele to half her sons.

Villager wisdom

- Here in Genesis Gardens we say Sertoli is the nursemaid and Leydig is the brewer. One feeds the young sperm, the other brews testosterone when LH knocks.
- Remember the garden calendar, traveler: estrogen builds the bed in the proliferative days, progesterone waters it in the secretory days, and when the corpus luteum withers, the bed is cleared away.
- Prolactin makes the milk and oxytocin moves the milk. Think P for produce and O for out the door.
- A seed in the wrong row is trouble. Most ectopic pregnancies sprout in the uterine tube, and the tube cannot stretch the way the uterus can.
- For X linked traits, a son gets his only X from his mother. So when the boys in a family are color blind, look first to the mother's side of the garden.

Diagrams

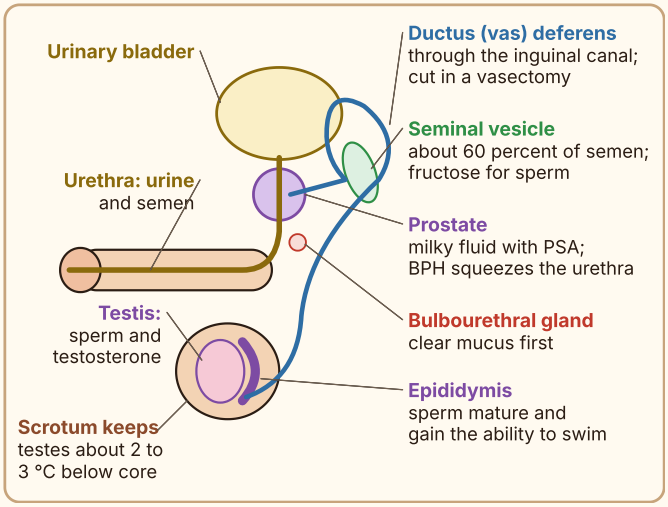
Testis, spermatogenesis and hormones



Meiosis I makes secondary spermatocytes; meiosis II makes spermatids; spermiogenesis remodels spermatids into sperm

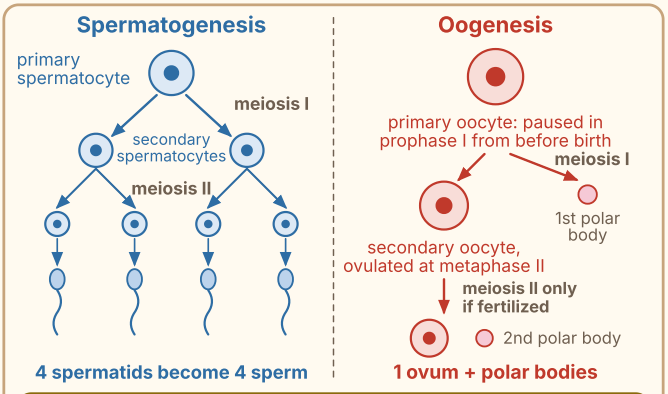
Sertoli cells nourish developing sperm and form the blood testis barrier; Leydig cells between the tubules make testosterone. FSH acts on Sertoli cells and LH on Leydig cells, and testosterone and inhibin feed back to slow GnRH, LH and FSH, which is why anabolic steroids shrink the testes.

Male reproductive system



Sperm made in the testis mature in the epididymis, travel up the ductus deferens through the inguinal canal and join secretions from the seminal vesicles, prostate and bulbourethral glands. The scrotum holds the testes a few degrees below core temperature, which sperm production needs.

Meiosis: sperm versus eggs

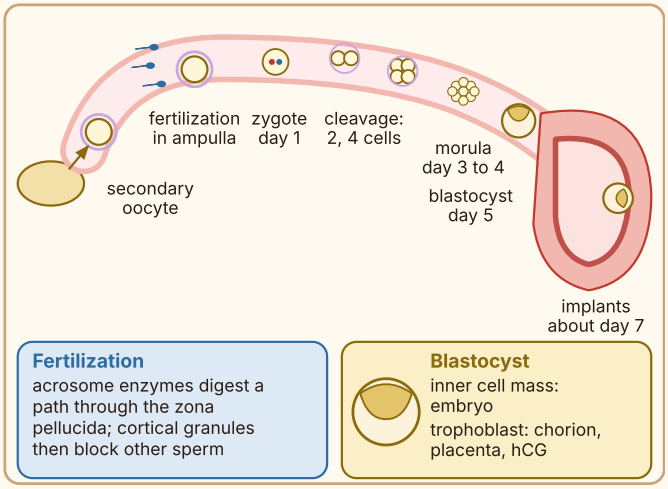


Meiosis: 2 divisions, crossing over, 4 haploid (n = 23) cells, all genetically different. Mitosis: 1 division, 2 identical diploid cells

Body cells: 46 chromosomes = 22 pairs of autosomes plus 1 pair of sex chromosomes (XX or XY)

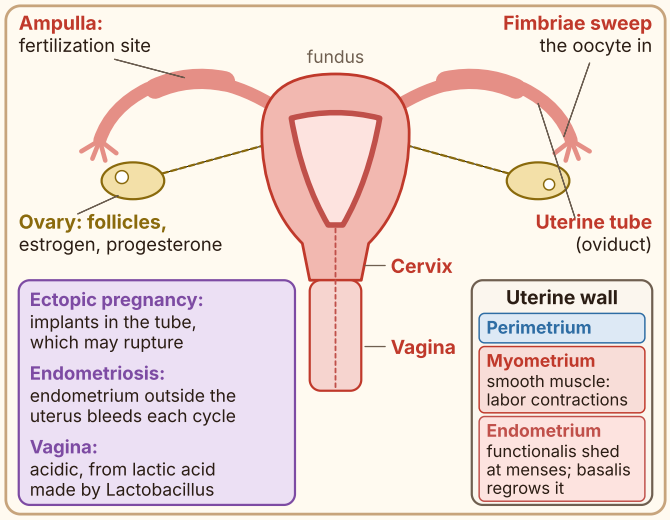
One primary spermatocyte yields four sperm, but one primary oocyte yields a single ovum plus polar bodies. The secondary oocyte is ovulated arrested in metaphase II and finishes meiosis only if a sperm enters.

From fertilization to implantation



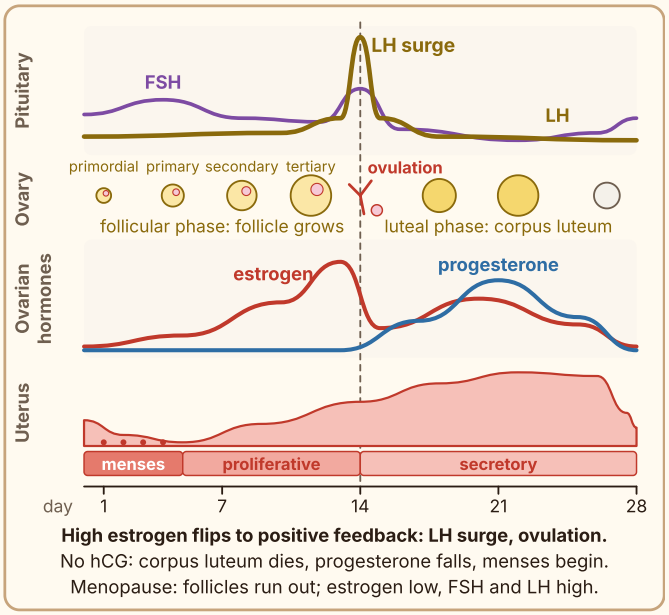
Fertilization usually happens in the ampulla; cleavage then divides the zygote into smaller and smaller cells, forming a morula and then a blastocyst that implants about a week after fertilization. The inner cell mass becomes the embryo and the trophoblast forms the chorion and placenta and makes hCG.

Female reproductive system



The secondary oocyte is swept into the uterine tube by the fimbriae and is usually fertilized in the ampulla. The myometrium is the smooth muscle that contracts in labor, and the stratum functionalis of the endometrium is shed at menstruation.

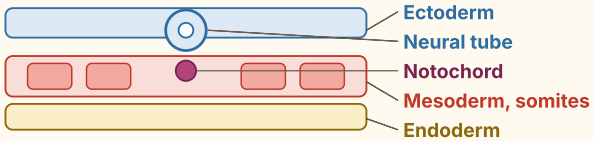
Ovarian and uterine cycles



High estrogen flips to positive feedback: LH surge, ovulation.
No hCG: corpus luteum dies, progesterone falls, menses begin.
Menopause: follicles run out; estrogen low, FSH and LH high.

Rising estrogen from the growing follicle rebuilds the endometrium; when estrogen is high it switches to positive feedback and the LH surge triggers ovulation. Progesterone from the corpus luteum readies the lining, and when the corpus luteum dies without hCG, progesterone falls and menstruation begins.

Germ layers and what they become



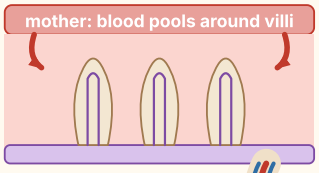
Notochord: induces the neural tube, later the nucleus pulposus

Ectoderm	Mesoderm	Endoderm
nervous system epidermis, hair, nails tooth enamel lens of the eye	heart, blood, vessels bone, cartilage skeletal muscle kidneys, gonads	lining of the GI and airways liver, pancreas thyroid bladder lining

weeks 1 to 2	weeks 3 to 8: embryo	week 9 on: fetus
pre embryonic	organs form; most at risk	growth, maturing

Ectoderm forms the nervous system and epidermis, mesoderm forms the heart, blood, bone and muscle, and endoderm forms the linings of the gut and airways. Organs form in weeks 3 to 8, the embryonic period, so that is when the embryo is most sensitive to harmful drugs.

Placenta and circulation at birth



Chorionic villi:
fetal capillaries; the two bloods do not mix

Umbilical vein (1): O2 rich,
placenta to fetus

Arteries (2): O2 poor,
fetus to placenta

hCG
from the trophoblast (chorion); keeps the corpus luteum making progesterone in early pregnancy; it is what a pregnancy test finds

Fetal shunts at birth

Foramen ovale: RA to LA
closes when LA pressure rises above RA: fossa ovalis

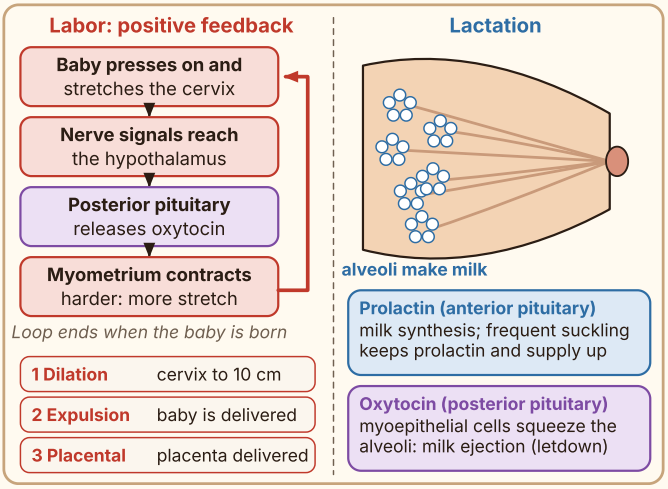
Ductus arteriosus:
pulmonary trunk to aorta; constricts: ligamentum arteriosum. If it stays open, blood shunts aorta to trunk

Ductus venosus:
bypasses the liver; becomes ligamentum venosum

Mother
blood volume and cardiac output both rise a lot

The single umbilical vein carries oxygen rich blood from the placenta to the fetus; maternal and fetal blood exchange across the villi without mixing. At the first breaths left atrial pressure rises above right atrial pressure and closes the foramen ovale, and the ductus arteriosus constricts.

Labor and lactation



Labor is positive feedback: stretch of the cervix releases oxytocin, which strengthens contractions and causes more stretch until birth. Prolactin drives milk synthesis in the alveoli and frequent suckling keeps it flowing, while oxytocin causes milk ejection.

Punnett squares and X linked inheritance

Autosomal: Aa × Aa

from dad \ from mom	A	a
A	AA	Aa
a	Aa	aa

1 AA : 2 Aa : 1 aa (genotypes)
3 dominant : 1 recessive look
1 in 4 (25 percent) show aa

X linked: carrier × normal

father \ mother	X^B	X^b
X^B	X^BX^B	X^BX^b
Y	X^BY	X^bY

daughters: half are carriers
sons: half are affected
b = recessive allele

Genotype: the alleles (AA, Aa, aa). Phenotype: the trait you see.
Homozygous: two identical alleles. Heterozygous: two different.
Aa × aa: half the children are aa (50 percent).
Males have one X, so one recessive allele shows. Affected
father × normal mother: all daughters carriers, no sons affected.

Two Aa carriers have a 1 in 4 (25 percent) chance of an aa child who shows the recessive trait. For an X linked recessive trait a carrier mother passes the allele to half her sons, who show it because males have only one X.

Self check

1. **An organism's observable traits, such as eye color, make up its:**
    A. Phenotype
    B. Genotype
    C. Karyotype
    D. Allele
2. **If the ductus arteriosus stays open after birth, blood will tend to shunt:**
    A. From the aorta into the pulmonary trunk
    B. From the pulmonary trunk into the aorta
    C. From the right atrium to the left atrium
    D. From the umbilical vein to the IVC
3. **A patient with endometriosis has pelvic pain that worsens around each menstrual period. This cyclic pattern occurs because ectopic endometrial tissue:**
    A. Is malignant tissue invading nearby organs
    B. Responds to ovarian hormones each cycle
    C. Produces its own LH surge
    D. Lacks estrogen receptors
4. **A color blind man (X linked recessive) has children with a woman homozygous for normal vision. Which prediction is correct?**
    A. Half of their sons are color blind
    B. All daughters are carriers; no sons are affected
    C. All daughters are color blind
    D. All sons are carriers
5. **Sertoli cells secrete inhibin when sperm count is high. Inhibin acts to:**
    A. Reduce LH release only
    B. Reduce GnRH and FSH release
    C. Increase testosterone output
    D. Increase FSH release
6. **The proliferative phase of the uterine cycle coincides with which ovarian phase and hormone?**
    A. Follicular phase; rising estrogen
    B. Luteal phase; rising progesterone
    C. Follicular phase; rising progesterone
    D. Luteal phase; falling estrogen
7. **The normally acidic pH of the vagina is maintained largely by:**
    A. Acidic secretions from the cervix
    B. Alkaline fluid from the seminal vesicles
    C. Lactobacillus bacteria producing lactic acid
    D. Estrogen secreted by the vaginal wall

8. **Fertilization most often takes place in which part of the female reproductive tract?**
    A. Body of the uterus
    B. Fimbriae of the uterine tube
    C. Cervix
    D. Ampulla of the uterine tube
9. **Unlike mitosis, meiosis:**
    A. Has one division and yields diploid cells
    B. Yields two genetically identical cells
    C. Occurs in body cells for tissue growth
    D. Has two divisions and yields haploid cells
10. **Ovulation is triggered by a sudden surge of which hormone?**
    A. Follicle stimulating hormone (FSH)
    B. Progesterone
    C. Prolactin
    D. Luteinizing hormone (LH)
11. **Clinicians may give synthetic oxytocin to strengthen labor. Once labor is underway, why do contractions keep intensifying on their own?**
    A. Rising progesterone stimulates the myometrium
    B. Cervical stretch triggers more oxytocin release
    C. Negative feedback shuts off oxytocin
    D. Prolactin causes uterine contraction
12. **Milk is synthesized by cells lining which mammary gland structures?**
    A. Lactiferous sinuses
    B. Alveoli
    C. Areolar glands
    D. Suspensory ligaments

Answer key

1. **A.** Phenotype. Phenotype is the physical expression of traits. Genotype is the tempting choice, but it refers to the allele combination that underlies those traits.
2. **A.** From the aorta into the pulmonary trunk. Once the lungs open, pulmonary pressure falls below aortic pressure, so flow through an open ductus reverses and runs from the aorta to the pulmonary trunk. Pulmonary trunk to aorta is the fetal direction, which depended on high fetal lung resistance.
3. **B.** Responds to ovarian hormones each cycle. Endometrial tissue outside the uterus still thickens and bleeds in step with estrogen and progesterone, causing inflammation and pain. Endometriosis is not cancer, so calling it malignant invasion is incorrect.
4. **B.** All daughters are carriers; no sons are affected. Every daughter receives the father's affected X plus a normal X from her mother, making her a carrier; sons receive his Y and a normal maternal X. Half of sons being affected would require a carrier mother.
5. **B.** Reduce GnRH and FSH release. Inhibin feeds back on the hypothalamus and anterior pituitary to lower GnRH and FSH, slowing spermatogenesis. Lowering LH alone is the tempting choice, but LH is regulated mainly by testosterone feedback.

6. **A.** Follicular phase; rising estrogen. Growing follicles secrete estrogen, which rebuilds the stratum functionalis after menses. Progesterone dominates later, during the luteal phase and secretory phase.
7. **C.** Lactobacillus bacteria producing lactic acid. Resident Lactobacillus species ferment glycogen from vaginal epithelial cells into lactic acid, which discourages pathogens. Seminal vesicle fluid is alkaline, so it raises rather than lowers vaginal pH.
8. **D.** Ampulla of the uterine tube. Sperm usually meet the secondary oocyte in the wide ampulla of the uterine tube. The uterus is where the blastocyst later implants, not where fertilization normally happens.
9. **D.** Has two divisions and yields haploid cells. Meiosis I separates homologous pairs and meiosis II separates sister chromatids, producing haploid gametes that differ genetically. Producing two identical cells describes mitosis, which is used for growth and repair.
10. **D.** Luteinizing hormone (LH). A midcycle LH surge causes the mature follicle to rupture and release the secondary oocyte. FSH also rises at that time, but LH is the hormone whose surge triggers ovulation.
11. **B.** Cervical stretch triggers more oxytocin release. The fetus pressing on the cervix sends signals that increase oxytocin release, which strengthens contractions and causes more stretch, a positive feedback loop ended by birth. Negative feedback would dampen, not amplify, contractions.
12. **B.** Alveoli. Glandular alveoli secrete milk, which drains through lactiferous ducts and collects in lactiferous sinuses near the nipple. The sinuses are tempting, but they only store milk rather than produce it.

Kingdom of Corpus · Dr. Akash Garg, Dr. Garg Learning · Aligned to OpenStax Anatomy and Physiology 2e · CC BY SA 4.0